MIL-A-47073(OS)
10 May 1974

SUPERSEDING
MIS 15296
18 February 1965

MILITARY SPECIFICATION

ADHESIVE, PLASTIC RESIN

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of plastic resin adhesives.

1.2 Classification. The two adhesive systems shall be classified as follows:

a. Type I, iron filled.

b. Type II, aluminum filled.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

STANDARDS

Military

MIL-STD-129 Marking for Shipment and Storage

(Copies of specifications, standards, publications, and drawings required by suppliers in connection with a specific procurement function should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein.

FSC 8040

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Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

ASTM D 638	Test for Tensile Properties of Plastics
ASTM D 695	Test for Compressive Properties of Rigid Plastics, Method of
ASTM D 790	Standard Method of Test for Flexural Properties of Plastics
ASTM D 1002	Test for Strength Properties of Adhesives in Shear by Tension Loading (Metal-to-Metal)
ASTM D 1084	Test for Consistency of Adhesives
ASTM D 1875	Test for Density of Adhesives in Fluid Form
ASTM D 2240	Test for Indentation of Rubber by Means of a Durometer, Method of

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

Uniform Classification Committee

Uniform Freight Classification Rules

(Application for copies should be addressed to Uniform Classification Committee, 202 Chicago Union Station, Chicago, Illinois, 60606).

American Trucking Association Publication

National Motor Freight Classification Rules and Container Regulations

(Application for copies should be addressed to the National Classification Board, 1424 Sixteenth, N.W., Washington, D.C. 20013)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

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3. Requirements

3.1 Preproduction sample. Unless otherwise specified (see 6.1), a preproduction sample shall meet the requirements of this specification.

3.2 Material. Type I and II adhesives shall be a two-component system consisting of plastic resins and reinforcing agents.

3.3 Physical properties. The physical properties of the adhesive components shall be as specified in Table I.

Table I

Physical Properties

Property	Type I, Requirements & Values	Type II, Requirements & Values	Test Method
Plastic Resin			
Weight per gallon at 24 to 26 C (75.2 to 78.8 F)	13.2 lbs, min	12.5 lbs, min	4.6.2
Viscosity at 22 to 23 C (71.6 to 73.4 F)	2,000,000 cps, min	2,000,000 cps, min	4.6.3
Color	black	metallic aluminum	4.6.1
Hardener			
Weight per gallon at 24 to 26 C (75 to 79 F)	8.0 lbs, min	8.0 lbs, min	4.6.2
Viscosity at 22 to 23 C (76 to 78 F)	20,000± 4,000 cps	20,000± 4,000 cps	4.6.3
Color	Off-white	Off-white	4.6.1

3.3.1 Adhesive system properties. The adhesive system properties of Type I and Type II shall be as specified in Table II.

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Table II

Adhesive System Properties

Property	Type I, Requirements & Values	Type II, Requirements & Values	Test Method
Hardness	75 ± 5 Barcol	70 ± 10 Barcol	4.6.4
Tensile Shear Strength			
2024 T3 Aluminum adherend	3100 psi, min	2500 psi, min	4.6.5
Half-hard cold-rolled steel adherend	2800 psi, min	2300 psi, min	
Tensile strength	6500 psi, min	5000 psi, min	4.6.6
Flexural strength	15000 psi, min	12000 psi, min	4.6.7
Compressive strength	18000 psi, min	14000 psi, min	4.6.8

3.4 Workmanship. The material furnished shall be uniform in quality and shall be free from impurities and other defects that could adversely affect its use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspections specified herein shall be classified as follows:

a. Preproduction inspections (see 4.4).

b. Quality conformance inspections (see 4.5).

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4.3 Inspection lot. An inspection lot shall consist of one batch of adhesive manufactured under essentially uniform conditions and are offered for inspection at any one time.

4.4 Preproduction inspection.

4.4.1 Preproduction sample inspections. Unless otherwise specified in the contract or purchase order, a sample of resin and hardener from the first lot manufactured under each contract or order shall be subjected to preproduction inspection.

4.4.2 Preproduction sampling and testing. One pint of resin and one pint of hardener shall be selected and subjected to the tests specified in Table III.

Table III

Preproduction Inspection

Examination or Test	Requirement Paragraph	Inspection Method Paragraph
Weight per gallon	3.3	4.6.2
Viscosity	3.3	4.6.3
Color	3.3	4.6.1
Hardness	3.3	4.6.4
Tensile shear strength	3.3	4.6.5
Tensile strength	3.3	4.6.6
Flexural strength	3.3	4.6.7
Compressive strength	3.3	4.6.8
Workmanship	3.4	4.6.1

4.4.3 Action in case of failure. No failures shall be allowed during the preproduction inspection. In case of failure, written approval from the procuring activity shall be obtained prior to submittal of subsequent lots for testing.

4.5 Quality conformance inspection. All production lots of adhesive offered for acceptance hereunder shall be subjected to quality conformance inspections.

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4.5.1 Quality conformance sampling. Sampling for visual examination shall be 100 percent of the lot. Failure of any sample to meet the requirements of 3.4 and Section 5 shall cause rejection of the sample. Sampling for all other examinations and tests shall be a representative sample of material selected at random. Failure of the sample to meet any other requirement of this specification shall cause rejection of the lot.

4.5.2 Procedure. Samples taken in accordance with 4.5.1 shall be subjected to the examinations and tests as specified in Table IV.

Table IV

Quality Conformance Inspection

Examination or Test	Requirement Paragraph	Inspection Method Paragraph
Visual examination	3.2, 3.4 and Section 5	4.6.1
Weight per gallon	3.3	4.6.2
Viscosity	3.3	4.6.3
Color	3.3	4.6.1
Hardness	3.3	4.6.4
Tensile shear strength 2024 T3 Aluminum adherend	3.3	4.6.5

4.6 Inspection methods.

4.6.1 Visual examination. Visually examine samples specified in 4.5.1 to determine compliance with color requirement (sec 3.3). Visually examine all containers in the lot to determine compliance with 3.2, 3.4, and Section 5.

4.6.2 Weight per gallon. Weight per gallon shall be determined in accordance with ASTM D 1875.

4.6.3 Viscosity. Viscosity shall be determined in accordance with ASTM D 1084 with the following exception:

A RVF Brookfield Viscosimeter operating at 22 to 24 degrees Celsius (C) (71.6 to 75.2 degrees Fahrenheit (F)) shall be utilized as follows:

	Spindle Number	Speed
Plastic Resin	7	2 rpm
Hardener	6	20 rpm

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4.6.4 Hardness. Hardness shall be determined in accordance with ASTM D 2240 with the following exceptions:

a. The mixture ratio shall be two parts by volume of plastic resin and one part by volume of hardener or three portions of plastic resin to one portion of hardener by weight.

b. The cure schedule for this test shall be 66 degrees C (150 degrees F) for 4 hours.

4.6.5 Tensile shear strength. Tensile shear strength shall be determined in accordance with ASTM D 1002. The preparation of test specimens shall be as follows:

a. Degrease surface to the bonded with methyl ethyl ketone or equivalent.

b. Clean surface with chromic acid solution at 66 to 77 degrees C (150 to 170 degrees F) for 5 to 10 minutes. The cleaning solution is prepared as follows:

10 parts by weight sodium dichromate
50 parts by weight 96 percent sulfuric acid
340 parts by weight distilled water

c. Rinse thoroughly with water and air dry.

d. The metal shall be 1/16 x 1 x 4 inches.

e. Overlap 1/2 inch and bond and cure as specified in 4.6.4.

f. Determine shear strength using cross head speed of 9.950 inches per minute.

4.6.6 Tensile strength. Tensile strength shall be determined in accordance with ASTM D 638 with exceptions as noted in 4.6.4.

4.6.7 Flexural strength. Flexural strength shall be determined in accordance with ASTM D 790 with exceptions as noted in 4.6.4.

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4.6.8 Compressive strength. Compressive strength shall be determined in accordance with ASTM D 695 with exceptions as noted in 4.6.4

5. PREPARATION FOR DELIVERY

5.1 Packaging and Packing. Unless otherwise specified, packaging and packing shall be Level C.

5.1.1 Level C Packaging. Material shall be packaged to afford adequate protection against deterioration and damage during shipment from the supply source to the first receiving activity.

5.1.2 Level C Packing. Packages shall be secured and packed in shipping containers in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rating. Containers shall comply with Uniform Freight Classification Rules or National Motor Freight Rules as applicable.

5.2 Marking. Each contanier shall be marked in accordance with MIL-STD-129. Marking shall include, but not be limited to, the following information:

a. Manufacturer's name and location.

b. Material grade name.

c. Net weight or volume.

d. Lot number, batch number, and date of manufacture.

e. Shelf life or storage limitations.

f. Number, title, and revision letter of this specification.

g. Toxicity/Fire prevention cautions, if needed.

h. Flash point.

6. NOTES

6.1 Intended Use. This specification covers two types of plastic resin adhesives.

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6.2 Ordering data. Procurement documents should specify, but not be limited to, the following information:

a. Title, number and date of this specification.

b. Whether a preproduction sample is required (3.1).

c. Minimum lot size, if applicable.

d. Place of delivery.

e. Size of container

f. Disposition of test data.

6.3 Supersession data. This specification includes the requirements of Missile Interim Specification MIS-15296, dated 18 February 1965.

Preparing Activity
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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